Lecture Notes on the IEEE 9-Bus AC Power-Flow Problem

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Abstract

The IEEE 9-bus system is a compact benchmark for the steady-state AC power-flow problem. It is small enough to inspect by hand, but it still contains the main modeling features that make power flow nontrivial: multiple generator buses, voltage magnitude control, nonlinear active- and reactive-power balance equations, branch losses, and reactive-power exchange with transmission lines. These notes formulate the problem, describe the specific 9-bus network used in this repository, discuss the solved operating point recorded in the checked-in MATPOWER-derived fixtures, and offer an economic interpretation for readers whose primary background is markets rather than power engineering.

1 Context and Relevance

Power-flow analysis is the basic steady-state calculation behind many power-system engineering tasks. Given a network model and specified generation, load, and voltage-control data, the AC power-flow problem computes the complex voltage at each bus and the real and reactive power flows on each branch. The result is not a dynamic trajectory; it is a snapshot of an operating point under balanced, sinusoidal, three-phase conditions.

This calculation matters because many operational and planning decisions depend on quantities that are not specified directly in the input data: voltage magnitudes at load buses, bus voltage angles, slack-bus real-power output, generator reactive-power output, branch flows, and system losses. A candidate dispatch may satisfy a real-power energy balance in aggregate while still producing unacceptable voltage profiles, excessive line flows, or reactive-power requirements. AC power flow is therefore the entry point for contingency analysis, voltage-security assessment, optimal power flow, state estimation validation, and many simulation studies.

It is also the physical feasibility model behind many nodal and security-constrained wholesale market calculations. Locational marginal prices (LMPs) are the dual variables on nodal active-power balance constraints in optimal power flow. Multipliers on binding branch-flow constraints produce congestion components of LMPs and settlement congestion rents, but the rents are not themselves dual variables. The slack-bus convention fixes the angle reference. Many production-grade market engines use a linearized *DC* power-flow approximation of the equations developed here, while AC power flow is the more complete network model those approximations are checked against. Understanding the AC equations on a small, transparent case is therefore useful background for anyone working with electricity-market models.

The case used here is the WSCC 3-machine 9-bus system. WSCC stands for Western Systems Coordinating Council, the historical reliability coordination body for the western North American interconnected power system; the benchmark is often called the Anderson 9-bus after its appearance in Anderson and Fouad's *Power System Control and Stability*. It is useful pedagogically because it is small but not degenerate. It has three generator buses (one slack, two PV), six PQ buses, and nine branches; impedances are reported on a 100 MVA system base. The topology is a meshed transmission network rather than a radial feeder, and the solution exhibits both real-power losses and significant reactive-power effects from line charging.

2 A Glossary for Readers from Economics

This note assumes familiarity with linear algebra and Newton's method but not with power engineering. The vocabulary is the main barrier. The following one-line definitions cover the terms that appear repeatedly. An economic analogue is given where one is genuinely close.

Bus. A node in the network — physically, a substation busbar. Read as "node" throughout.

Branch. An edge between two buses. May be a transmission line or a transformer. Read as "arc" or "link."

Per-unit (p.u.). A non-dimensionalization. Every quantity is divided by a reference base value, so a voltage of 1.00 p.u. means "the nominal design voltage at this bus," and a power of 1.00 p.u. means 100 MW on the system used here (100 MVA base). Treat it exactly like reporting macro variables as fractions of GDP — the base is sometimes implicit but always there.

Real power (MW). The energy actually delivered per unit time; what consumers pay for and what generators are dispatched on.

Reactive power (MVAr). A network-support quantity needed to sustain AC voltages in the presence of inductive and capacitive elements. It does not deliver useful work in the same sense as real power, but transmission cannot operate without it. Market treatment varies by jurisdiction: reactive support may be covered by interconnection requirements, cost-based compensation, tariff obligations, or ancillary-service procurement rather than a nodal price analogous to LMP.

Slack bus. The bus where voltage magnitude and angle are fixed by convention and the generator output is whatever the rest of the problem implies. It is analogous to a numéraire only in the narrow sense that one reference is fixed: here one angle is fixed at zero, and all other angles are then *relative* to it.

PV bus. A generator bus with specified real-power output and voltage magnitude (the "P" and the "V"). Reactive output and angle are solved. In this fixture, reactive limits are not enforced, so the solved Q_G is whatever the unconstrained power-flow equations require to hold the voltage setpoint.

PQ bus. A bus with specified real and reactive demand (the "P" and the "Q"). Both voltage magnitude and angle are solved. Read as "a passive consumption node."

Voltage angle (θ_i). A scalar at each node. Differences in θ across a branch drive real-power flow on that branch. The closest economic analogy is to a relative-gradient variable: angle differences, like price differences, help determine the direction and quantity of exchange, but they are physical state variables rather than market prices.

Series impedance. The branch parameters r and x. Resistance r produces real-power losses (the "transaction cost" interpretation); reactance x governs how strongly an angle difference pushes real power across the branch.

Shunt charging (b). The total branch charging susceptance in the MATPOWER branch model. The model places $jb/2$ at each end of the branch; for positive capacitive charging, this contributes reactive injection that depends on the local voltage magnitude.

Tap ratio, phase shift. Transformer settings. Both equal 1 and 0 respectively in this fixture, so they can be ignored here.

Generator step-up transformer (GSU). The transformer that connects a generator to the high-voltage transmission grid. The IEEE 9-bus has three of them (branches 1–4, 3–6, 8–2).

A useful one-liner, with the analogy kept deliberately loose: *angle differences drive real-power exchange, real power is the commodity, reactive power is voltage support, losses are the transaction cost, and the slack bus supplies the reference angle.*

3 Mathematical Formulation

3.1 Network Model

Let the system have n buses and complex bus-voltage phasors

$$V_i = |V_i| e^{j\theta_i}, \qquad i = 1, \dots, n.$$

The nodal admittance matrix is

$$Y_{\text{bus}} = G + jB,$$

where G_{ik} and B_{ik} are the conductance and susceptance entries of the bus admittance matrix. The series-admittance part is Laplacian-like, with weights set by branch impedances; line charging, shunts, transformer taps, and phase shifts add the details that make the full Y_{bus} more than a plain graph Laplacian. At each bus, the net complex power injection is

$$S_i = P_i + jQ_i = V_i I_i^*,$$

where P_i is real power and Q_i is reactive power. Using $I = Y_{\text{bus}} V$, the standard polar-form AC power-flow equations are

$$P_i = |V_i| \sum_{k=1}^{n} |V_k| \left[G_{ik} \cos(\theta_i - \theta_k) + B_{ik} \sin(\theta_i - \theta_k) \right], \tag{1}$$

$$Q_i = |V_i| \sum_{k=1}^{n} |V_k| \left[G_{ik} \sin(\theta_i - \theta_k) - B_{ik} \cos(\theta_i - \theta_k) \right]. \tag{2}$$

The specified net injections are formed from positive generation and positive load:

$$P_i^{\text{spec}} = P_{G,i} - P_{D,i}, \qquad Q_i^{\text{spec}} = Q_{G,i} - Q_{D,i}.$$

All real and reactive powers in these notes are reported on the 100 MVA base used by the fixture files unless explicitly shown in MW or MVAr.

3.2 Bus Types and Unknowns

The power-flow equations are solved with different specifications at different bus types:

- **Slack (reference) bus**: specifies $|V|$ and θ. Its real and reactive generation are outputs of the solution. The slack bus serves two structural roles. First, it provides the angle reference $\theta_1 = 0$, without which the system is invariant under a global phase shift and the equations are underdetermined — the reference-angle role. Second, its real-power output absorbs the active-power residual caused by losses once the non-slack real-power injections are fixed. Reactive power is handled differently: the slack bus and all PV buses have solved reactive outputs, so reactive balance is not assigned to a single bus.

- **PV bus**: specifies real-power generation P_G and voltage magnitude $|V|$. The voltage angle θ and the reactive generation Q_G are both solved. It is therefore wrong to read the case-side Q_G at a PV bus as a setpoint; in the fixtures it is just an initial value that the solver overwrites.

- **PQ bus**: specifies real and reactive load. Both $|V|$ and θ are solved.

The resulting nonlinear system is square: there are $n-1$ active-power equations (one per non-slack bus) and $n - n_{\text{PV}} - 1$ reactive-power equations (one per PQ bus). It is solved by Newton's method on the mismatch residual; the standard derivation, including the four Jacobian sub-blocks, is sketched in Appendix A.

4 The Specific IEEE 9-Bus Network

The fixture used here is `data/case9.json`, with solved values from `data/case9_solution.json`. The case has three generator buses: bus 1 is the slack bus, while buses 2 and 3 are PV buses. Loads are located at buses 5, 7, and 9. The total load is

$$P_D = 315.00 \text{ MW}, \qquad Q_D = 115.00 \text{ MVAr}.$$

The case-side P_G at the slack bus and Q_G at all generator buses are MATPOWER initial values rather than scheduled targets, so it is the solved generator outputs that should be compared against the load. Only P_G at the two PV buses (163.00 and 85.00 MW) is a true setpoint, and the solver honors both exactly.

Table 1: Bus data and solved voltages and generation for the IEEE 9-bus case.

Bus	Type	P_D MW	Q_D MVAr	P_G^{case} MW	Q_G^{case} MVAr	$\lvert V\rvert$ p.u.	θ deg	P_G^{sol} MW	Q_G^{sol} MVAr
1	slack	0.00	0.00	72.30	27.03	1.0400	0.0000	71.64	27.05
2	PV	0.00	0.00	163.00	6.54	1.0250	9.2800	163.00	6.65
3	PV	0.00	0.00	85.00	-10.95	1.0250	4.6648	85.00	-10.86
4	PQ	0.00	0.00	0.00	0.00	1.0258	-2.2168	0.00	0.00
5	PQ	90.00	30.00	0.00	0.00	1.0127	-3.6874	0.00	0.00
6	PQ	0.00	0.00	0.00	0.00	1.0324	1.9667	0.00	0.00
7	PQ	100.00	35.00	0.00	0.00	1.0159	0.7275	0.00	0.00
8	PQ	0.00	0.00	0.00	0.00	1.0258	3.7197	0.00	0.00
9	PQ	125.00	50.00	0.00	0.00	0.9956	-3.9888	0.00	0.00

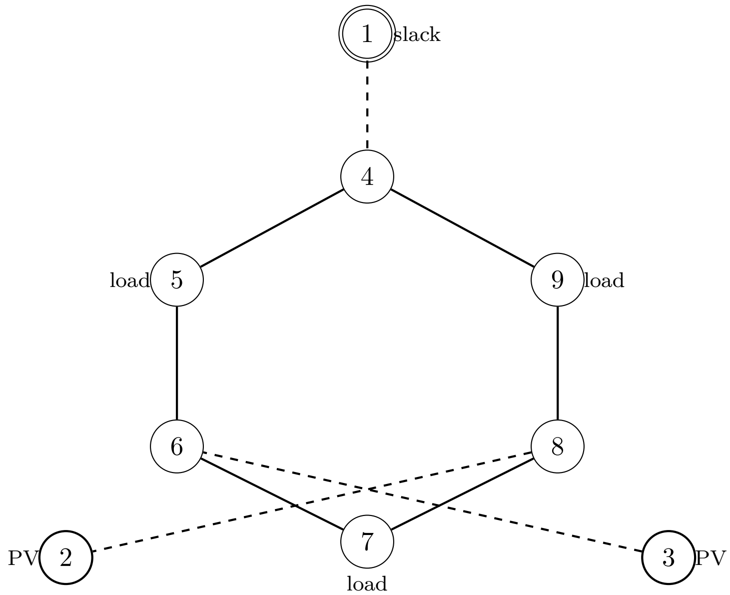

Figure 1: Topology of the IEEE 9-bus (WSCC) system. The double-circle node is the slack bus; thick-circle nodes are PV generator buses; "load" annotations mark the three PQ buses with non-zero demand. Dashed lines (1–4, 3–6, 8–2) are the generator step-up transformer branches; solid lines are transmission branches. The interior loop 4−5−6−7−8−9−4 is the meshed sub-network through which generator power redistributes.

Table 2: Branch parameters for the 9-bus case. All series and charging parameters are in p.u. on the system base; `tap_ratio` = 1.0 and `phase_shift` = 0 for every branch in this fixture, so those columns are omitted. Branches with $r = 0$ and $b = 0$ (1–4, 3–6, 8–2) are the generator step-up transformers in the original WSCC model; the fixture stores them as plain reactances because per-unit base voltages are not tracked here.

Branch	From	To	r	x	b
1	1	4	0.00000	0.05760	0.00000
2	4	5	0.01700	0.09200	0.15800
3	5	6	0.03900	0.17000	0.35800
4	3	6	0.00000	0.05860	0.00000
5	6	7	0.01190	0.10080	0.20900
6	7	8	0.00850	0.07200	0.14900
7	8	2	0.00000	0.06250	0.00000
8	8	9	0.03200	0.16100	0.30600
9	9	4	0.01000	0.08500	0.17600

Topologically, the meshed sub-network is the loop

$$4 \to 5 \to 6 \to 7 \to 8 \to 9 \to 4.$$

Each generator feeds into this loop through a single radial branch: gen 1 via branch 1–4, gen 2 via branch 8–2, and gen 3 via branch 3–6. On the radial hops there is no choice of path, but once power enters the loop it divides according to the network impedances and the bus voltage-angle profile. This is the structural reason the angle profile matters: it is the state that determines how much real power takes each direction around the loop.

5 Discussion of the Solved Operating Point

5.1 Active-power balance and losses

The solved generator outputs are

$$P_G^{\text{sol}} = 319.641 \text{ MW},$$
$$Q_G^{\text{sol}} = 22.840 \text{ MVAr}.$$

Since the total real load is 315.000 MW, the real-power loss is

$$P_{\text{loss}} = 319.641 - 315.000 = 4.641 \text{ MW},$$

which equals the sum of $P_f + P_t$ over all branches as it must. The slack bus supplies 71.641 MW; the PV buses hold their scheduled real-power outputs of 163.000 MW at bus 2 and 85.000 MW at bus 3.

5.2 Voltage profile

The voltage magnitudes remain close to nominal. The highest solved voltage is the slack-bus voltage, 1.0400 p.u.. The lowest voltage is at bus 9, 0.9956 p.u., which is also the largest load bus (125 MW and 50 MVAr). The drop is driven primarily by the local reactive demand: increasing reactive load Q_D tends to lower voltage at heavily loaded buses, while increasing net reactive injection tends to raise it. Bus 9 has both the largest Q_D and two relatively impedance-heavy branches connecting it to the rest of the network.

A subtler observation is that bus 6 sits at 1.0324 p.u., which is *above* the 1.0250 p.u. PV setpoint used at generator buses 2 and 3. Bus 6 is directly adjacent to generator bus 3, and it also sits in a meshed region with non-negligible line charging and nearby voltage control. The solved branch-end quantities show mixed local reactive exchange at bus 6, so the safest interpretation is not that one line "causes" the higher voltage by itself, but that the network solution allows an interior PQ-bus voltage to rise above a generator voltage setpoint. This is why the PV-setpoint values cannot be read as upper bounds on the system voltage profile.

5.3 Angles and active-power flow

The voltage angles span from $-3.9888°$ at bus 9 to $9.2800°$ at bus 2. In an AC transmission network, active-power transfer is strongly tied to angle differences across each branch through the near-lossless approximation

$$P_{ij} \;\approx\; \frac{|V_i|\,|V_j|}{x_{ij}}\,\sin(\theta_i - \theta_j).$$

Branch 8–2 illustrates this clearly. The fixture stores the orientation as 8 to 2, with $P_f = -163.000$ MW and $P_t = 163.000$ MW, so the real-power flow is from bus 2 toward bus 8. Plugging in $|V_2| = 1.0250$, $|V_8| = 1.0258$, $x_{8\text{-}2} = 0.0625$, and $\theta_2 - \theta_8 = 9.2800° - 3.7197° = 5.5603°$:

$$P_{28} \approx \frac{1.0250 \times 1.0258}{0.0625} \sin(5.5603°) = 1.6303 \text{ p.u.} = 163.03 \text{ MW},$$

which matches the solved 163.00 MW to three figures. The lossless approximation is essentially exact on this branch because $r = 0$. Reading the table this way — always check angle differences when interpreting branch flows — is the most useful intuition the 9-bus case offers.

5.4 Reactive-power balance

Table 3: Solved branch powers. The "from" and "to" quantities follow the fixture branch orientation. Positive endpoint power is injected from that bus terminal into the branch; negative endpoint power is delivered from the branch into that bus terminal.

From	To	P_f MW	Q_f MVAr	P_t MW	Q_t MVAr	P_ℓ MW	Q_ℓ MVAr
1	4	71.641	27.046	-71.641	-23.923	0.000	3.123
4	5	30.704	1.030	-30.537	-16.543	0.166	-15.513
5	6	-59.463	-13.457	60.817	-18.075	1.354	-31.531
3	6	85.000	-10.860	-85.000	14.955	0.000	4.096
6	7	24.183	3.120	-24.095	-24.296	0.088	-21.176
7	8	-75.905	-10.704	76.380	-0.797	0.475	-11.502
8	2	-163.000	9.178	163.000	6.654	0.000	15.832
8	9	86.620	-8.381	-84.320	-11.313	2.300	-19.694
9	4	-40.680	-38.687	40.937	22.893	0.258	-15.794

The sign discussion generalizes: on branch 7–8, the negative P_f indicates that real power enters the branch at bus 8 and leaves at bus 7, even though the fixture orientation runs the other way.

The reactive-power column tells a different story. Summing $Q_f + Q_t$ over all branches gives exactly

$$\sum_{\text{branches}} (Q_f + Q_t) = Q_G^{\text{sol}} - Q_D = 22.840 - 115.000 = -92.160 \text{ MVAr}.$$

This is not a coincidence and not a violation of conservation; it is the reactive analogue of the active-loss identity, but with a sign that can go either way. Several branches inject more reactive power through their shunt charging elements than they absorb in their series reactances, so the per-branch reactive endpoint sum can be negative. In this case the line charging dominates: the network as a whole is a net reactive-power *source*, supplying 92.160 MVAr that the generators do not have to provide. This is one of the reasons AC power flow must model reactive power explicitly rather than treating it as a small correction to a real-power-only calculation.

5.5 Economic interpretation

For readers approaching this from electricity markets rather than power engineering, four observations on the same numbers are worth pulling out explicitly.

Losses create a physical wedge that can enter prices. The 4.641 MW of real losses is generation that has to be procured but that no consumer pays for at a meter. In OPF and market models that represent marginal losses, those losses enter the LMP decomposition as a loss component alongside energy and congestion components. The 4.641 MW value here is the system-wide physical loss at this operating point, not itself a price component. Losses are a real economic cost; they just do not appear on the demand side of the energy-balance constraint.

Reactive power is voltage support, not a second energy product. The 92.160 MVAr that the network supplies from line charging is, in effect, an uncontrolled capacitive injection in this model rather than a generator dispatch decision. The 22.840 MVAr from generators is a solved operating-point quantity. In real systems generator reactive output can consume capability and may be compensated through tariffs, requirements, or ancillary-service mechanisms, but this fixture has no generator capability curves or reactive-power cost data. The economic lesson is therefore limited: reactive support is necessary and scarce in operations, but these numbers alone do not imply a marginal cost.

Angle differences are the physical state behind DC flow. The DC power-flow approximation, which underlies many nodal market and planning models, replaces the AC equations with a linearization in which $P_{ij} \propto (\theta_i - \theta_j)/x_{ij}$. The angles are primal state variables in that model, while LMPs are the dual variables on the nodal balance constraints. Worked example: branch 8–2 carries 163 MW at an angle wedge of 5.56°, exactly tracking the lossless formula. In a DC-OPF analogue, price separation would come from binding constraints or modeled marginal losses, not from the mere existence of a nonzero angle difference. The structural insight from the AC calculation is therefore physical first: real power is driven by angle differences; prices respond when those physical flows interact with economic costs and constraints.

Bus 9 is the weakest-voltage load bus. Bus 9 has the lowest voltage (0.9956 p.u.) and the most negative angle ($-3.99°$). It is a large-load corner of the loop connected through branches 8–9 and 9–4; it is not topologically isolated from the slack injection point, but its local load and branch impedances make it the lowest-voltage bus in this solution. In the LMP picture, those same physical factors can contribute to price separation when they increase marginal losses or make transmission constraints bind. A power-flow solution alone does not determine which bus has the highest LMP; that requires an OPF objective, generator costs, and active constraints.

6 Takeaways

1. The IEEE 9-bus problem is a minimal but meaningful AC power-flow benchmark: it includes slack, PV, and PQ bus behavior in a small meshed network with three generator step-up transformers and a six-bus transmission loop.

2. The unknown state consists of voltage angles at non-slack buses and voltage magnitudes at PQ buses. Slack-bus real and reactive power and PV-bus reactive power are solved outputs, not specified inputs. The slack bus absorbs the active-power residual from losses; reactive balance is shared through the solved reactive outputs of the slack and PV buses. Case-side P_G at the slack and Q_G at every generator should be read as initial values, not setpoints.

3. The solved real-power generation exceeds total real load by 4.641 MW, which is the network real-power loss at this operating point.

4. Voltage angle differences govern active-power flows through the near-lossless relation $P_{ij} \approx (|V_i||V_j|/x_{ij})\sin(\theta_i - \theta_j)$. Branch 8–2 reproduces its solved 163 MW flow from this formula to three figures.

5. Branch-flow signs must be read relative to the stored branch orientation. A negative endpoint power means power is delivered from the branch into that bus terminal rather than injected from the bus into the branch.

6. Reactive-power behavior is qualitatively different from real-power loss accounting because line charging can inject reactive power. In this case the network is a net reactive-power source of 92.160 MVAr, and an interior PQ bus (bus 6) sits above the generator PV setpoint.

7. The vocabulary has useful but limited market analogies: angle differences drive real-power exchange, real power is the commodity, reactive power is voltage support, losses are transaction costs, and the slack bus fixes the reference angle. AC power flow is the feasibility layer beneath nodal OPF and LMP calculations.

A Newton–Raphson Solution

The non-slack bus active-power mismatches and the PQ-bus reactive-power mismatches are

$$\Delta P_i = P_i^{\text{spec}} - P_i(V,\theta), \quad i \notin \{\text{slack}\},$$

$$\Delta Q_i = Q_i^{\text{spec}} - Q_i(V,\theta), \quad i \in \{\text{PQ}\}.$$

With n_{PV} PV buses, the unknowns are the $n-1$ non-slack voltage angles and the $n - n_{\text{PV}} - 1$ PQ-bus voltage magnitudes, giving a square nonlinear system. Let J_{calc} be the Jacobian of the calculated injections,

$$J_{\text{calc}} = \begin{bmatrix} \dfrac{\partial P}{\partial \theta} & \dfrac{\partial P}{\partial |V|} \\ \dfrac{\partial Q}{\partial \theta} & \dfrac{\partial Q}{\partial |V|} \end{bmatrix} = \begin{bmatrix} H & N \\ M & L \end{bmatrix},$$

restricted to the rows and columns corresponding to the unknowns above. Because the mismatch is defined as specified minus calculated injection, the Newton step solves

$$J_{\text{calc}} \begin{bmatrix} \Delta\theta \\ \Delta|V| \end{bmatrix} = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix},$$

then adds the step to the current voltage angles and magnitudes. If one instead defines the residual Jacobian directly, it is $-J_{\text{calc}}$ and the right-hand side changes sign as well. Iteration continues until $\max_i(|\Delta P_i|, |\Delta Q_i|)$ falls below a tolerance, typically 10^{-6} to 10^{-8} p.u. The MATPOWER fixture used here does not record the iteration count or final mismatch, so we cite only the converged state.